# STACKS, DERIVED CATEGORIES OF SPACES: PERFECT-ENOUGH

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 6543-6551 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 6553-6581, the complete author proof (27 body lines).

*Editorial note.* Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 41 of queue_spaceperfect.json (pb=1146, stmt 6543-6551, proof 6553-6581).

*Editorial note.* stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

*Editorial note.* Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-perfect-enough' also exists in stacks-perfect — a different proposition in a different chapter file.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 08FA: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 790–804. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 08HP: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3786–3790. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $A$ be a ring. Let $X$ be an algebraic space over $A$ which is quasi-compact and quasi-separated. Let $K \in D^-_{QCoh}(\mathcal{O}_X)$. If $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is pseudo-coherent in $D(A)$ for every perfect $E$ in $D(\mathcal{O}_X)$, then $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is pseudo-coherent in $D(A)$ for every pseudo-coherent $E$ in $D(\mathcal{O}_X)$.*

*Date*: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* There exists an integer $N$ such that $R\Gamma(X, -) : D_{QCoh}(\mathcal{O}_X) \to D(A)$ has cohomological dimension $N$ as explained in Lemma 08FA. Let $b \in \mathbf{Z}$ be such that $H^i(K) = 0$ for $i > b$. Let $E$ be pseudo-coherent on $X$. It suffices to show that $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is $m$-pseudo-coherent for every $m$. Choose an approximation $P \to E$ by a perfect complex $P$ of $(X, E, m - N - 1 - b)$. This is possible by Theorem 08HP. Choose a distinguished triangle

$$P \to E \to C \to P[1]$$

in $D_{QCoh}(\mathcal{O}_X)$. The cohomology sheaves of $C$ are zero in degrees $\geq m - N - 1 - b$. Hence the cohomology sheaves of $C \otimes^{\mathbf{L}} K$ are zero in degrees $\geq m - N - 1$. Thus the cohomology of $R\Gamma(X, C \otimes^{\mathbf{L}} K)$ are zero in degrees $\geq m - 1$. Hence

$$R\Gamma(X, P \otimes^{\mathbf{L}} K) \to R\Gamma(X, E \otimes^{\mathbf{L}} K)$$

is an isomorphism on cohomology in degrees $\geq m$. By assumption the source is pseudo-coherent. We conclude that $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is $m$-pseudo-coherent as desired. $\square$